YERRAGUNTA KUSUMITA

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SUMMARY

CS undergraduate with hands-on experience building AI-driven and retrieval-based systems, including RAG pipelines, vector search, and NLP-based grounding, alongside software engineering experience at Samsung R&D Institute Bangalore. Proficient in Python, C++, Java, and Linux-based development.

EDUCATION

PES University, Bangalore, India **Expected 2027**
B.Tech. in Computer Science and Engineering CGPA: 8.34/10

TECHNICAL SKILLS

Programming: Python, C++, Java, C, SQL, JavaScript, Dart
ML/AI : RAG, Object Tracking (ByteTrack),CUDA, ONNX, FAISS, ChromaDB,LangChain, OpenCV, YOLOv8, PyTorch
Backend & Web: Flask, FastAPI, React, REST APIs
Backend Tools: Flask, FastAPI, Node.js, REST APIs, MySQL, Flutter, Tree-sitter

EXPERIENCE

Samsung R&D Institute Bangalore — *Developer Intern*, *Magpie* May 2026 – July 2026

- Designed an end-to-end FAISS-based semantic memory-indexing and retrieval system in a Linux-based Android build/test environment, surfacing similar historical failures for CTS root-cause analysis.
- Architected a multi-stage retrieval pipeline across 7 modules, integrating HAL source code, CTS documentation, and historical failure logs to localize failures to source methods and lines.
- Developed a Tree-sitter-based Java AST indexer to extract method metadata, signatures, and inter-method relationships for structured retrieval.
- Built a Node.js/Express REST bridge connecting Python scripts with an LLM-powered VS Code extension, supporting asynchronous task submission and status tracking entirely offline.
- Automated foundational root-cause localization for Camera HAL test failures, reducing manual logcat-to-source tracing.

DEEVIA Software India Private Limited — *Software Engineering Intern* June 2025 – July 2025

- Built a YOLOv8/OpenCV computer-vision pipeline for pedestrian and object detection on fisheye-camera video, targeting distortion-heavy edge cases.
- Consolidated 19 inconsistent and duplicate label classes into 11 standardized object classes across 100+ annotated images.
- Evaluated detection performance using precision, recall, and mAP across pedestrians, vehicles, traffic signs, and crosswalks.

PROJECTS

FlutterPulse – Gesture Engine Performance Optimization
Flutter, *Concurrency*, *Performance Engineering*

- Isolated compute-heavy gesture processing from the UI thread, reducing latency from approximately 200 ms to 20 ms under heavy load (90% improvement).

GritSense – Real-Time Volleyball Analytics & ML Performance Engineering
Python, *C++*, *PyTorch*, *TorchScript*, *ONNX*, *CUDA*

- Developed a GPU inference benchmark comparing PyTorch and ONNX execution using latency and throughput.
- Built a YOLOv8/ByteTrack detection and tracking pipeline with persistent multi-player tracking and IoU-matched pose skeleton overlays.
- Profiled video-processing stages to identify bottlenecks in preprocessing, inference, postprocessing, and data movement.
- Implemented and benchmarked CUDA C++ kernels, evaluating correctness and execution time against CPU baselines.

To-Do List App
Flutter, *Dart*, *Local Storage*

- Developed a cross-platform mobile application using Flutter and Dart, supporting task creation, status tracking, and persistent local storage.
- Implemented task-management functionality and local data persistence to retain tasks across app sessions.
- Built a mobile user interface designed for managing everyday tasks.

Hallucination-Aware Hybrid RAG System for Coffee Disease Diagnosis
Python, *RAG*, *Vector Search*, *Sentence Transformers*

- Combined structured JSON disease data and unstructured agricultural PDFs in a hybrid RAG pipeline to improve retrieval coverage across diseases.
- Added multi-turn clarification for ambiguous symptoms, improving Top-1 diagnostic accuracy from 50% to 66.67% and achieving 83.33% Top-3 accuracy.
- Implemented cross-response consistency checks (88.89% score; 25% hallucination rate) and Corrective RAG relevance filtering (Precision@5: 48.19%).

PUBLICATIONS & CERTIFICATIONS

Agro-Veritas: Hallucination-Aware Coffee Disease Diagnosis using Retrieval-Augmented Generation — Submitted to ICECCE 2026, Kuala Lumpur, Malaysia. (Under Review) | Paper
AWS Certification — Amazon EC2 & VPC, Jan 2026 | Certificate